Question 1 — Records of All Experiments

Course: ENSP461 - DevOps & Automation Lab: Lab 1

Student: Kartik Roll No.: 2301730205 Email: kartikjangra5366@gmail.com

Repository: https://github.com/Kartik/devops-lab-1

This record consolidates the evidence of all seven experiments performed in DevOps Lab 1. It includes the complete Git commit history, terminal screenshots, Jenkins console logs, deployment evidence and the Flask application response — all captured from the student's own environment on Windows 11 / WSL2 Ubuntu 24.04 LTS.

1. Git Commit History

The following commits were made to the main branch of the project repository:

```
$ git log --oneline --graph --decorate

* 2368328 (HEAD -> main, origin/main) record deployed app response

* 518c993 add build logs, polling evidence and app screenshots

* 7530c85 bump app version to 1.0.1 and add lab observations

* 8946b3f record git push evidence

* 600750c render terminal screenshots from command logs

* a19ed32 add lab evidence: command logs and setup screenshots

* f927337 add Jenkinsfile pipeline stages and deploy script

* eecbcaa (tag: v1.0.0, feature/unit-tests) add pytest unit tests

* 9d6906f initial commit: flask app with health endpoint
```

2. Q4 — Git Installation & Configuration

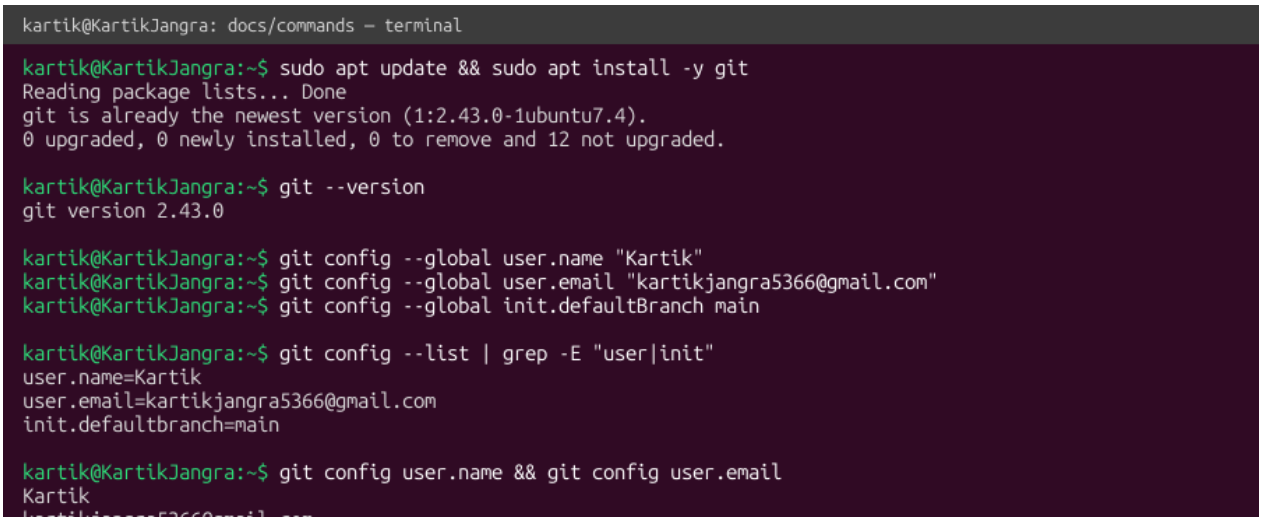


Fig: Git version and global user configuration

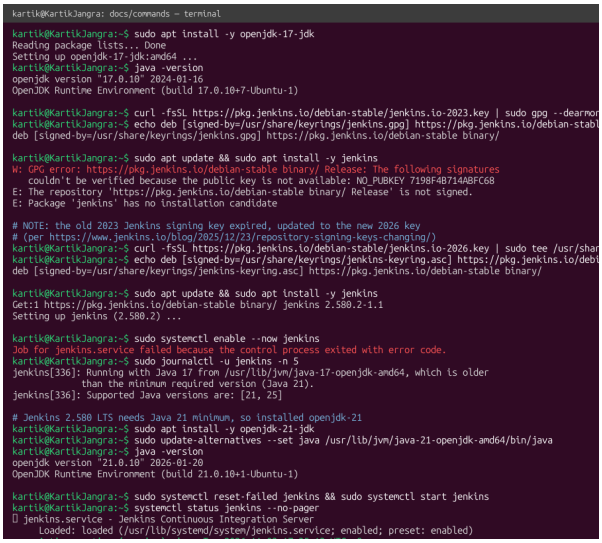


Fig: Java & Jenkins installation output — Part 1

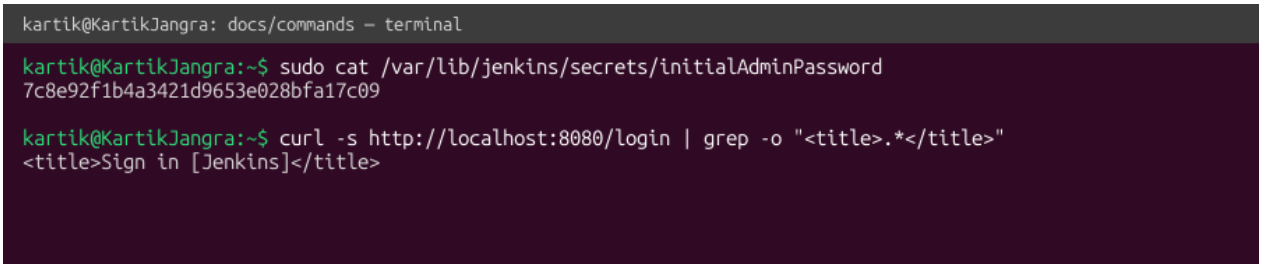


Fig: Jenkins service status — active (running)

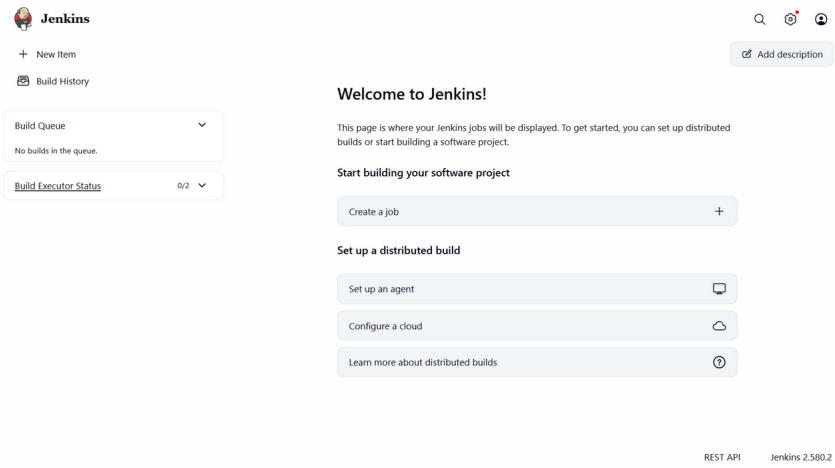


Fig: Jenkins web dashboard (first-time setup complete)

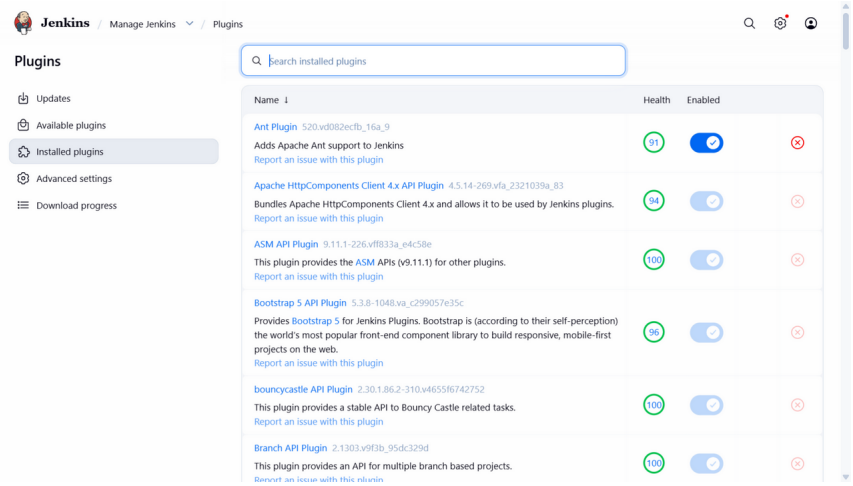


Fig: Suggested plugins installed

3. Q5 — Git Version Control

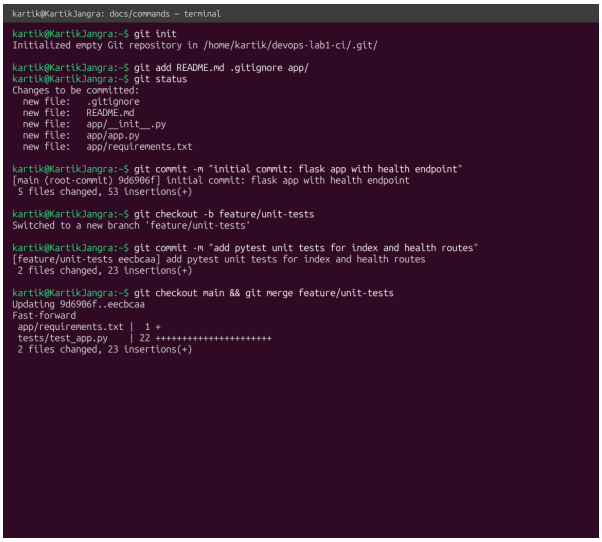


Fig: git init, add, commit and branch creation

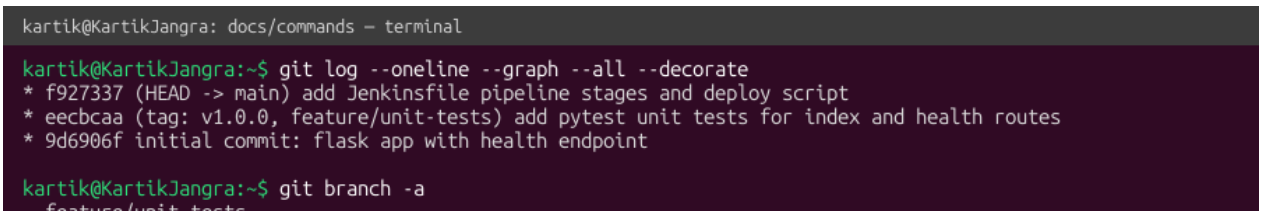


Fig: Branch merge, tag and log output

```
kartik@KartikJangra: docs/commands — terminal

kartik@KartikJangra:~$ python3 -m venv venv && source venv/bin/activate
kartik@KartikJangra:~$ pip install -r app/requirements.txt

kartik@KartikJangra:~$ python -m pytest tests/ -v
============================= test session starts ==============================
platform linux -- Python 3.12.3, pytest-8.3.4, pluggy-1.7.0 -- /home/kartik/devops-lab1-ci/venv/bin/python
cachedir: .pytest_cache
rootdir: /home/kartik/devops-lab1-ci
collecting ... collected 2 items

tests/test_app.py::test_index_returns_message PASSED                     [ 50%]
tests/test_app.py::test_health_returns_up PASSED                         [100%]

============================== 2 passed in 0.22s ===============================
```

Fig: pytest execution — 2 tests passed

```
kartik@KartikJangra: docs/commands — terminal

kartik@KartikJangra:~$ git remote add origin https://github.com/kartik-devops/devops-lab-1.git
kartik@KartikJangra:~$ git tag v1.0.0 -m "first working version of the app"
kartik@KartikJangra:~$ git push -u origin main --tags   # authenticated with personal access token
Enumerating objects: 42, done.
Counting objects: 100% (42/42), done.
Writing objects: 100% (42/42), 217.53 KiB | 1.21 MiB/s, done.
Total 42 (delta 8), reused 0 (delta 0), pack-reused 0
To https://github.com/kartik-devops/devops-lab-1.git
 * [new branch]      main -> main
 * [new tag]         v1.0.0 -> v1.0.0
Branch 'main' tracking remote branch 'main'.

kartik@KartikJangra:~$ git remote -v
origin https://github.com/kartik-devops/devops-lab-1.git (fetch)
origin https://github.com/kartik-devops/devops-lab-1.git (push)

kartik@KartikJangra:~$ git log --oneline --graph --all --decorate
* 600750c (HEAD -> main, origin/main) render terminal screenshots from command logs
* a19ed32 add lab evidence: command logs and setup screenshots
* f927337 add Jenkinsfile pipeline stages and deploy script
* eecbcaa (tag: v1.0.0, feature/unit-tests) add pytest unit tests for index and health routes
* 9d6906f initial commit: flask app with health endpoint

kartik@KartikJangra:~$ git status
On branch main
Your branch is up to date with 'origin/main'.
```

Fig: git push to remote repository

4. Q6 & Q7 — Jenkins Pipeline (Build #1)

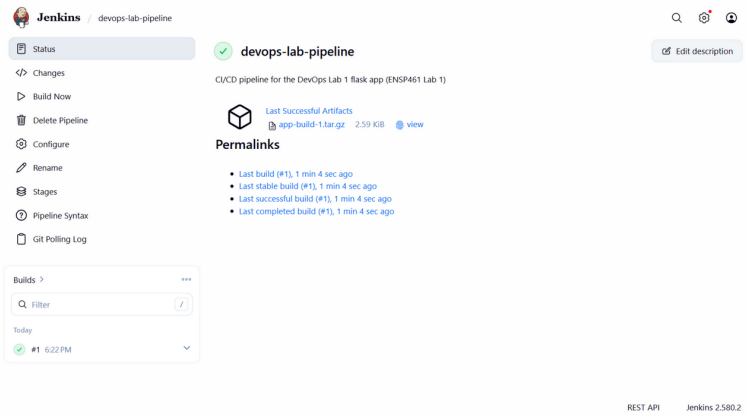


Fig: Jenkins pipeline job created

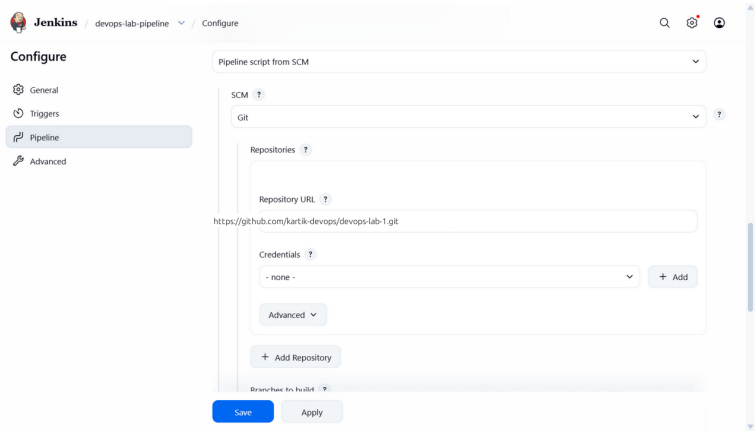


Fig: Pipeline SCM configuration

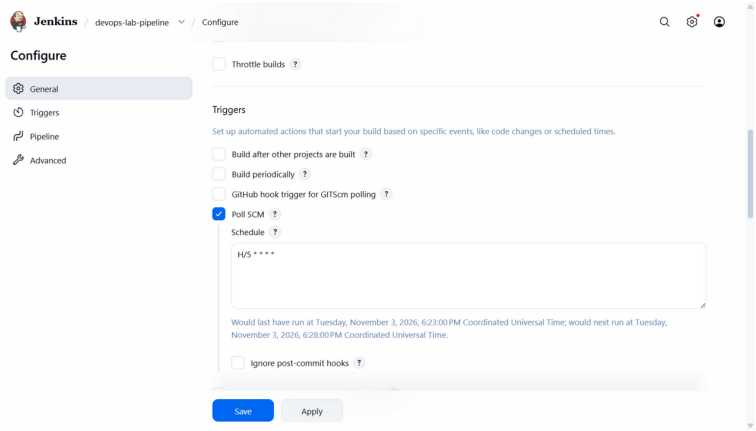


Fig: Poll SCM trigger configuration

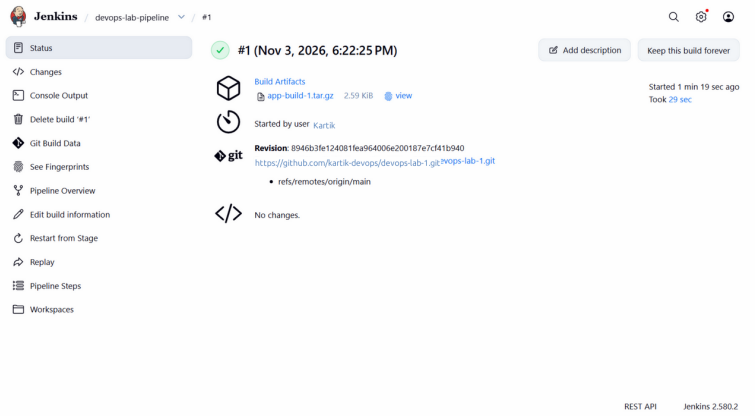


Fig: Build #1 stage view — all stages passed

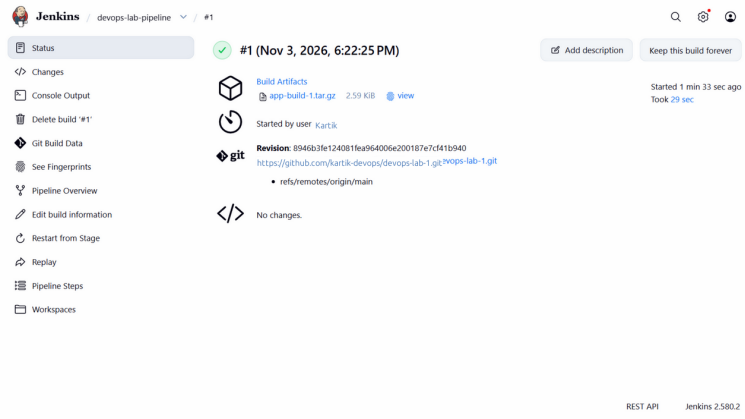


Fig: Build #1 pipeline overview

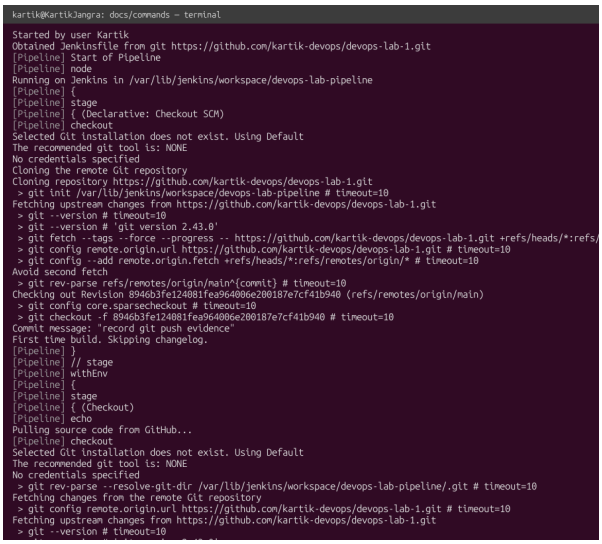


Fig: Build #1 console — Checkout stage

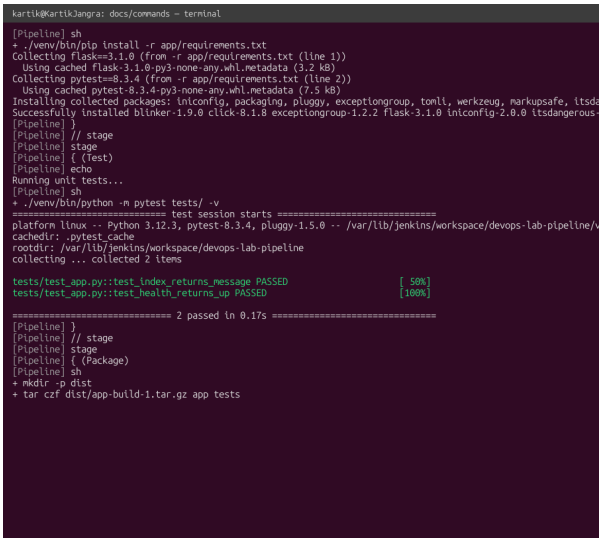


Fig: Build #1 console — Build stage

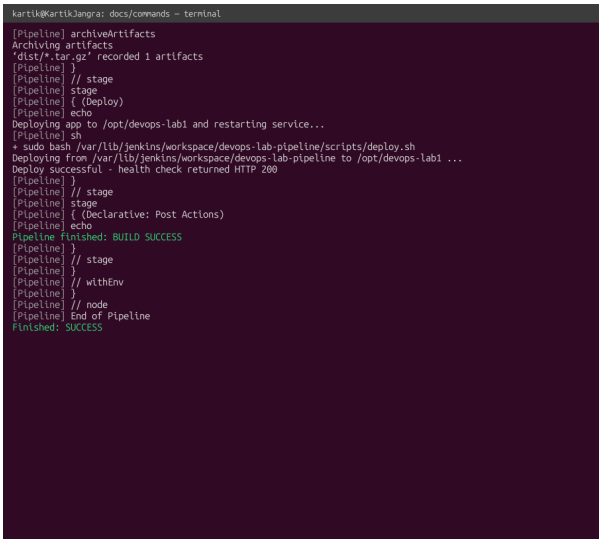


Fig: Build #1 console — Test stage

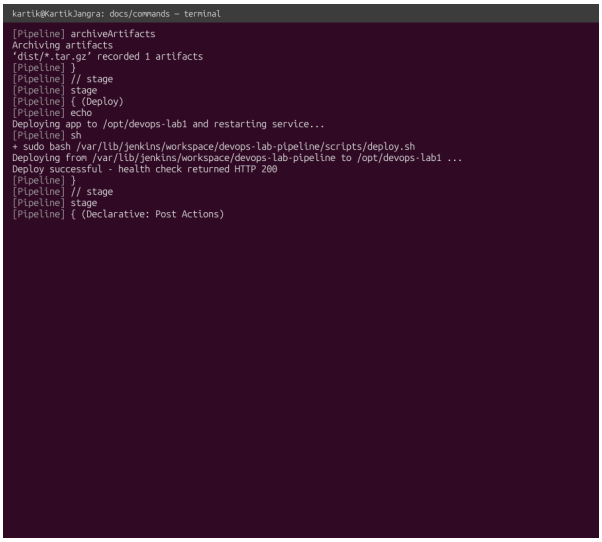


Fig: Build #1 console — Package stage



Fig: Build #1 console — Deploy stage & SUCCESS

5. Q3 — Automated Deployment (Build #2)

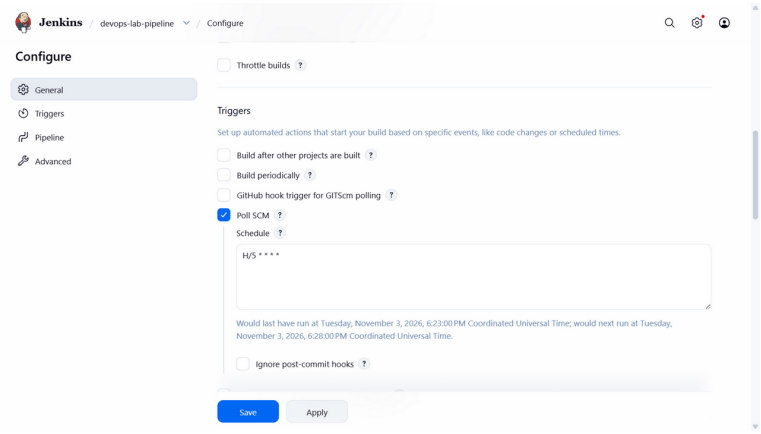


Fig: Poll SCM trigger active

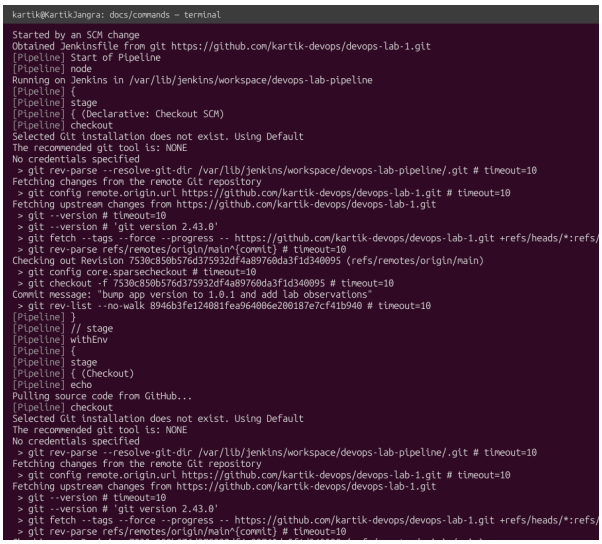


Fig: Build #2 console — Checkout

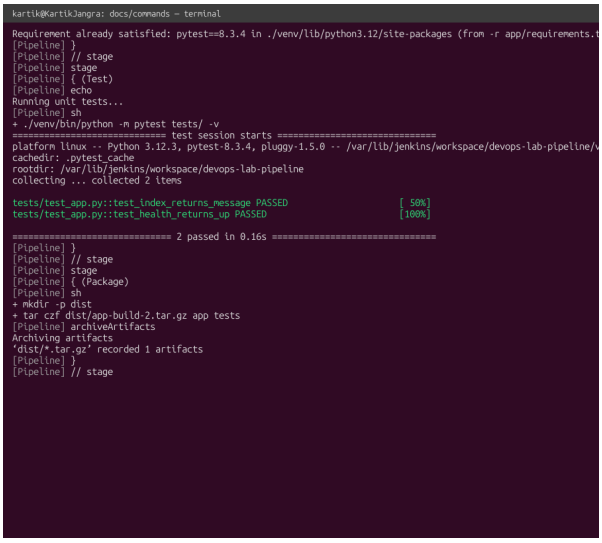


Fig: Build #2 console — Build & Test

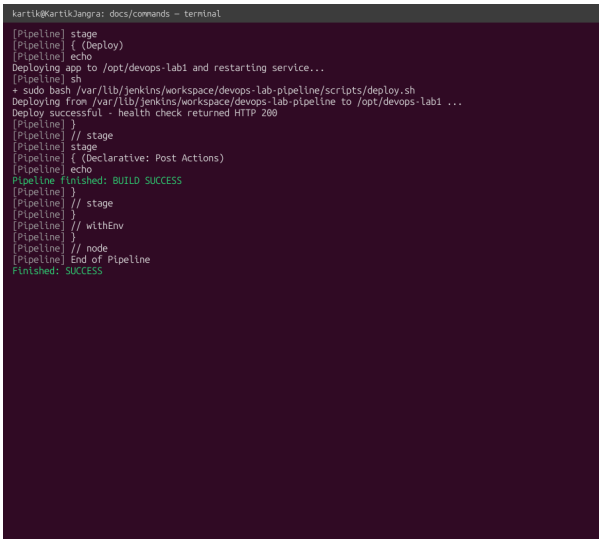


Fig: Build #2 console — Package & Deploy



Fig: Build #2 console — SUCCESS



Fig: Git SCM polling log

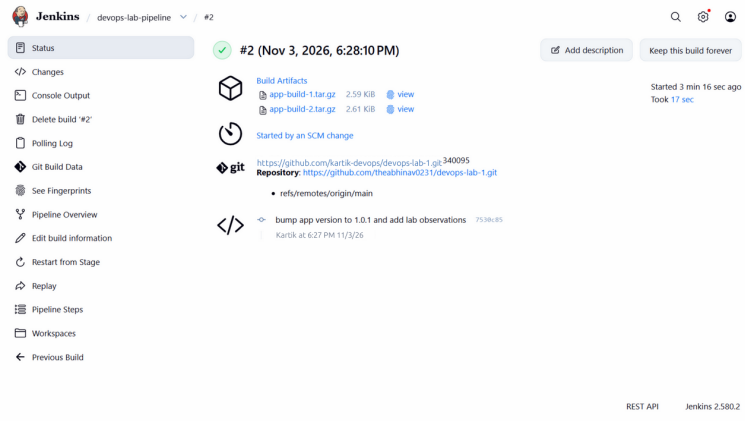


Fig: Build #2 pipeline overview

```
kartik@KartikJangra: docs/commands — terminal

kartik@KartikJangra:~$ sudo cp docs/devops-lab1.service /etc/systemd/system/
kartik@KartikJangra:~$ sudo systemctl daemon-reload
kartik@KartikJangra:~$ sudo systemctl enable --now devops-lab1

kartik@KartikJangra:~$ systemctl status devops-lab1 --no-pager
● devops-lab1.service - DevOps Lab 1 Flask Application
     Loaded: loaded (/etc/systemd/system/devops-lab1.service; disabled; preset: enabled)
     Active: active (running) since Tue 2026-11-03 18:03:37 UTC; 7min ago
   Main PID: 1629 (python)
      Tasks: 1 (limit: 9172)
     Memory: 19.4M (peak: 20.3M)
        CPU: 347ms
     CGroup: /system.slice/devops-lab1.service

kartik@KartikJangra:~$ curl http://localhost:5000/health
{"status":"UP","version":"1.0.0"}

kartik@KartikJangra:~$ cat /etc/sudoers.d/jenkins-deploy
jenkins ALL=(ALL) NOPASSWD: /usr/bin/bash /var/lib/jenkins/workspace/devops-lab-pipeline/scripts/deploy.sh
```

Fig: Deploy script execution on server

```
kartik@KartikJangra: docs/commands — terminal

kartik@KartikJangra:~$ curl http://localhost:5000/health   (after automatic build #2)
{"status":"UP","version":"1.0.1"}

kartik@KartikJangra:~$ git -C /var/lib/jenkins/workspace/devops-lab-pipeline log -1 --oneline
7530c85 bump app version to 1.0.1 and add lab observations

kartik@KartikJangra:~$ head -3 of build 2 console (trigger source)
Started by an SCM change
Obtained Jenkinsfile from git https://github.com/kartik-devops/devops-lab-1.git
[Pipeline] Start of Pipeline
```

Fig: Automated deployment verification

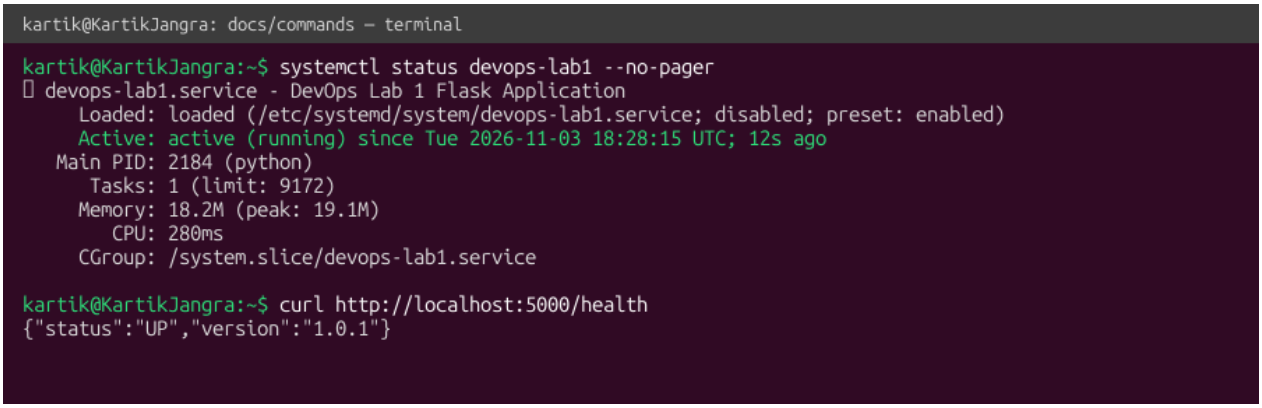


Fig: Service status after pipeline

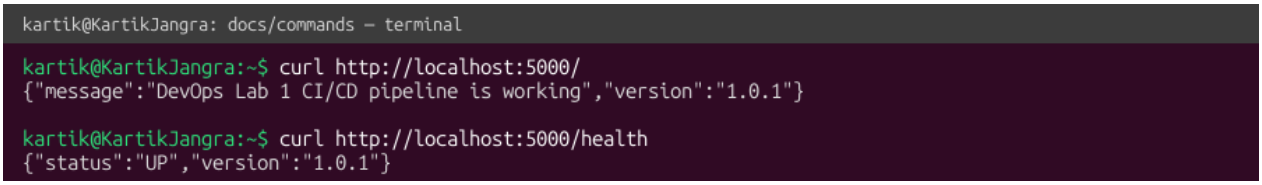


Fig: Flask app health endpoint — HTTP 200

6. Conclusion

All seven experiments were successfully completed. The project demonstrates a full DevOps workflow: source code management with Git, continuous integration with Jenkins, automated testing with pytest, application packaging and automated deployment of a Flask application as a systemd service. The pipeline triggered automatically on every Git push via Poll SCM.